

# Chapter 1

## TT

### 1.1 SYS-0

The alphabet is listed as follows.

- ◇  $V = \{v_i\}_{i \geq 0} \cup \{A_i\}_{i \geq 0}$ ;
- ◇  $C$  is the set of constant symbols;
- ◇  $B = \{\Pi, \lambda, \Sigma, \dots\}$  is the set of binders;
- ◇  $U = \{\text{Type}_i\}_{i \geq 0}$  is the set of universes;
- ◇  $P = \{(\cdot), \cdot, \cdot, \equiv\}$  is the set of punctuation.

All of these sets are at most countable.

The whole formulas we use is contained by set

$$F = V|C|U|B(V : F)F|(FF)|(CF^*).$$

**Notation 1.1.1** *Here we announce some commonly used notations for avoiding ambiguity.*

- ◇  $\mathbf{a} = a_0 a_1 \cdots a_m$ ;
- ◇  $f(\mathbf{x}) := f$  where  $f \in F$  and  $\text{Fr } f \subset \{x_0, \dots, x_m\}$ ;
- ◇  $(f\mathbf{x})$  refers  $(\cdots((fx_0)x_1)\cdots)$ , note that this may differ the notation  $(fx_0x_1 \cdots x_m)$ , which is equal to our first definition;
- ◇ for any  $b \in B$ ,  $b(\mathbf{v} : \mathbf{A})N := b(v_0 : A_0)b(v_1 : A_1) \cdots b(v_m : A_m)N$ ;
- ◇  $\mathbf{v} : \mathbf{A}$  refers
 
$$v_0 : A_0 \quad v_1 : A_1 \quad \cdots \quad v_s : A_s;$$
- ◇  $M \rightarrow N := \Pi(v : M)N$ , in which  $v \notin \text{Fr } N$ ;
- ◇ to distinguish between terms in  $F$  and constants in  $C$ , we use a mono font for constants;
- ◇  $\mathcal{U} := \text{Type}_{\aleph_0}$ , and  $M : \mathcal{U}$  implies there exists a  $k < \aleph_0$  such that  $M : \text{Type}_k$ .
- ◇ for any relation  $\mathcal{R} \subset F^2$ ,  $\text{Compatible}\mathcal{R}$  is a abbreviation of the statement: for all if  $MRN$ , then for any  $L, A, \phi_1, \dots, \phi_s \in F$ ,  $b \in B$ ,  $\mathbf{c} \in C$  and  $v \in V$ ,  $(ML)\mathcal{R}(NL)$ ,  $(LM)\mathcal{R}(LN)$ ,  $b(v : A)MRb(v : A)N$  and

$$(\mathbf{c}\phi_1 \cdots \phi_k M \phi_{k+1} \cdots \phi_s)\mathcal{R}(\mathbf{c}\phi_1 \cdots \phi_k N \phi_{k+1} \cdots \phi_s)$$

**Definition 1.1.2** A definition  $D$  has the form

$$\mathbf{c}(v : A) \equiv d(v) : B,$$

where  $v \in V$ ,  $\mathbf{c} \in C$  and all of  $A, B, d(v)$  are in  $F$ .

**Definition 1.1.3** A sequent has the form

$$D \quad v : A \quad M : A$$

where  $M : A$  is called a judgement, and both  $M$  and  $A$  are in  $F$ . The set of all sequents is denoted by  $\text{Sqt}$ .

**Legal judgements** consist a subset of  $F : F$ , which is derived from a series of **rules**. In our SYS-0, we first introduce the following rules, each of them can be viewed as a relation on  $\text{Sqt}^* \times \text{Sqt}$ .

### Sort

If  $m < n$ , then

$$\_ \_ \text{Type}_m : \text{Type}_n$$

### Var

If  $v \in V \setminus (\Gamma \cap V)$ , then

$$\begin{array}{l} D \quad \Gamma \quad A : \mathcal{U} \\ D \quad \Gamma \quad v : A \quad v : A \end{array}$$

### Weak

If  $v \in V \setminus (\Gamma \cap V)$ , then

$$\begin{array}{l} D \quad \Gamma \quad M : A \\ D \quad \Gamma \quad B : \mathcal{U} \\ D \quad \Gamma \quad v : B \quad M : A \end{array}$$

### Compu

Given the equivalence relation  $=_a \subset F^2$ , if  $B =_a B'$ , then

$$\begin{array}{l} D \quad \Gamma \quad M : B \\ D \quad \Gamma \quad B' : \mathcal{U} \\ D \quad \Gamma \quad M : B' \end{array}$$

## 1.2 SYS-1

我们通过引入新符号, 增加新规则和关系来增加新的项和类型, 这个过程大致遵循一个固定的工作流: 1. Formation: 引入新规则, 让系统在指定条件下可以导出新类型; 2. Constuction: 引入新规则, 让系统在指定条件下可以导出新项, 并且这些项都具有新类型; 3. Elimination: 引入新规则, 告知系统新项应该如何使用; 4. Computaiton: 引入  $F$  上的新关系, 告知系统新系统中某些表达和旧系统中的某些表达具有同样的语义.

现在我们将拓展 SYS-0, 增加  $\lambda$ -Terms 和  $\Pi$ -Types.

### SYS-1: Form

If  $s_1 \leq s_2$ , then

$$\begin{array}{l} D \quad \Gamma \quad A : \mathcal{U} \\ D \quad \Gamma \quad v : A \quad B : \mathcal{U} \\ D \quad \Gamma \quad \Pi(v : A)B : \mathcal{U} \end{array}$$

**SYS-1: Constr**

$$\begin{array}{ll}
D \quad \Gamma \quad v : A & M : B \\
D \quad \Gamma \quad \Pi(v : A)B : \mathcal{U} \\
D \quad \Gamma \quad \lambda(v : A)M : \Pi(v : A)B
\end{array}$$

**SYS-1: Elim**

$$\begin{array}{ll}
D \quad \Gamma \quad M : \Pi(v : A)B \\
D \quad \Gamma \quad N : A \\
D \quad \Gamma \quad (MN) : B \frac{N}{v}
\end{array}$$

**SYS-1: Compu**

**Definition 1.2.1**  $\rightarrow_0 \subset F^2$  is the samllest relation satisfies the following properties:

◇ for any  $A, M$  in  $F$ ,  $b \in B$  and  $v_0, v_1 \in V$  such that  $v_1 \in (M \cap V) \setminus \{v_0\}$

$$b(v_0 : A)M \rightarrow_0 b(v_1 : A)M \frac{v_1}{v_0};$$

◇ Compatible  $\rightarrow_0$ .

**Definition 1.2.2**  $\rightarrow_1 \subset F^2$  is the samllest relation satisfies the following properties:

◇ for any  $A, M, N$  in  $F$ ,  $(\lambda(v : A)MN) \rightarrow_1 M \frac{N}{v}$ ;

◇ Compatible  $\rightarrow_1$ .

$=_1$  is the equivalence genertated by  $\bigcup \{\rightarrow_i\}_{i=0}^1$ .

**1.3 SYS-2**

在 SYS-1 的基础上加入  $\Sigma$ -Types 得到 SYS-2.

**SYS-2: Form**

$$\begin{array}{ll}
D \quad \Gamma \quad A : \mathcal{U} \\
D \quad \Gamma \quad v : A & B : \mathcal{U} \\
D \quad \Gamma \quad \Sigma(v : A)B
\end{array}$$

**SYS-2: Constr**

$$\begin{array}{ll}
D \quad \Gamma \quad \Sigma(v : A)B \\
D \quad \Gamma \quad a : A \\
D \quad \Gamma \quad b : B \frac{a}{v} \\
D \quad \Gamma \quad (\text{pair}ab) : \Sigma(v : A)B
\end{array}$$

定义 Elim 前先明确需求, 对任何 Type 或者 Universe  $Q$ , 通过给定的项  $\phi : \Pi(v : A)(Bv \rightarrow Q)$ , 其中  $Bv$  指的是公式的合并  $(Bv)$ , 我们可以构造出一个具有类型  $(\Sigma(v : A)Bv) \rightarrow Q$  的项  $\psi$ . 所以, 有理由将规则设计成下面这个样子:

$$\begin{array}{ll}
D \quad \Gamma \quad \Sigma(v : A)B : \mathcal{U} \\
D \quad \Gamma \quad \phi : \Pi(v : A)(Bv \rightarrow Q) \\
D \quad \Gamma \quad (\text{Elim}Q\phi) : (\Sigma(v : A)Bv) \rightarrow Q
\end{array}$$

然后再加入新关系  $\rightarrow_2$ , 其包含下面内容

$$((\text{Elim}Q\phi)(\text{pair}ab)) \rightarrow_2 ((\phi a)b)$$

除此之外还有一种更普遍的方法, 它是从更复杂的功能中下放到这里的.

**SYS-2: Elim**

$$\begin{array}{l}
D \quad \Gamma \quad \Sigma(v : A)B : \mathcal{U} \\
D \quad \Gamma \quad (\mathbf{Ind}\Sigma(v : A)B) : \Pi(X : (\Sigma(v_0 : A)B) \rightarrow \mathcal{U}). \\
\quad \quad \quad [\Pi(v_1 : A)(v_2 : Bv_2)(X\mathbf{pair}_{v_1v_2}) \rightarrow \Pi(v_3 : \Sigma(v_4 : A)Bv_4)(Xv_3)]
\end{array}$$

可见如果给出常函数  $H = \lambda(X : \Sigma(v_0 : A)B)Q : \Pi(X : (\Sigma(v_0 : A)B) \rightarrow \mathcal{U})$ , 还有

$$\phi : \Pi(v_0 : A)(v_1 : Bv_0)H(\mathbf{pair}_{v_0v_1}) = \Pi(v_0 : A)(v_1 : Bv_0)Q,$$

那么

$$(\mathbf{Ind}\Sigma(v : A)B)(H\phi) : (\Sigma(v : A)Bv) \rightarrow Q.$$

**SYS-2: Compu**

**Definition 1.3.1**  $\rightarrow_2 \subset F^2$  is the samllest relation satisfies the following properties:

◇ for all  $a, b, A, B, \phi, H \in F$ ,

$$(\mathbf{Ind}\Sigma(v : A)B)(H\phi ab) \rightarrow_2 (\phi a)b$$

◇ Compatible  $\rightarrow_2$ .

$=_2$  is the equivalence genertated by  $\bigcup\{\rightarrow_i\}_{i=0}^2$ .

**1.4 SYS-3**

在 SYS-2 中加入自然数类型  $\mathbb{N}$  使其成为 SYS-3.

**SYS-3: Form**

$$\begin{array}{l}
D \quad \Gamma \quad \bar{\_} : \mathcal{U} \\
D \quad \Gamma \quad \bar{\mathbb{N}} : \mathcal{U}
\end{array}$$

**SYS-3: Constr**

$$\begin{array}{l}
D \quad \Gamma \quad \mathbb{N} : \mathcal{U} \\
D \quad \Gamma \quad \mathbf{0} : \mathbb{N}
\end{array}$$

and

$$\begin{array}{l}
D \quad \Gamma \quad \mathbb{N} : \mathcal{U} \\
D \quad \Gamma \quad \mathbf{S} : \mathbb{N} \rightarrow \mathbb{N}
\end{array}$$

**SYS-3: Elim**

$$\begin{array}{l}
D \quad \Gamma \quad \mathbb{N} : \mathcal{U} \\
D \quad \Gamma \quad (\mathbf{Ind}\mathbb{N}) : \Pi(X : \mathbb{N} \rightarrow \mathcal{U})[(X\mathbf{0}) \rightarrow \Pi(v_0 : \mathbb{N})((Xv_0) \rightarrow X(\mathbf{S}v_0)) \rightarrow \Pi(v_1 : \mathbb{N})(Xv_1)]
\end{array}$$

**Example 1.4.1** Let  $P$  be  $\lambda(v : \mathbb{N})\text{Type}_0$ , which implies that  $P$  is a property on  $\mathbb{N}$ . In addition, assume that  $p_0$  and  $p_S$  are two legal elements in  $F$ , satisfy  $p_0 : P\mathbf{0}$  and  $p_S : \Pi(v : \mathbb{N})((Pv) \rightarrow P(\mathbf{S}v))$ , means that both propositions  $P\mathbf{0}$  and  $\forall n : \mathbb{N} Pn \rightarrow P\mathbf{S}n$  are ture, then

$$(\mathbf{Ind}\mathbb{N})(Pp_0p_S) : \Pi(v : \mathbb{N})(Pv),$$

namely, the proposition  $\forall n : \mathbb{N} P(n)$  is true.

**SYS-3: Compu**

**Definition 1.4.2**  $\rightarrow_3 \subset F^2$  is the samllest relation satisfies the following properties:

◇ for all  $\phi, a, b, c \in F$ ,

$$(\mathbf{IndN})(\phi ab\theta) \rightarrow_3 a$$

and

$$(\mathbf{IndN})(\phi ab(\mathbf{Sc})) \rightarrow_3 (bc)((\mathbf{IndN})(\phi abc))$$

◇ Compatible  $\rightarrow_3$ .

$=_3$  is the equivalence genertated by  $\bigcup \{\rightarrow_i\}_{i=0}^3$ .

**1.5 SYS-CIC**

上述流程推广后就得到 SYS-CIC.

**Definition 1.5.1** For any  $t \in F$ , we have the following definitions:

◇ an Arity is generated by the syntax

$$\mathbb{A} = U|\Pi(V : F)\mathbb{A};$$

◇ Strictly Positive Types of  $t$  is the set

$$\mathbb{P}(t) = \{\Pi(v_0 : B_0) \cdots (v_r : B_r)[(\cdots (tN_0) \cdots N_s)] | B_i, N_i \in F \wedge t \notin B_i, N_i\};$$

in fact, it is a set defined inductively:

$$\begin{aligned} G_0 &= \{t\} \\ G_{n+1} &= \{(xN) | x \in G_n \wedge N \in F \wedge t \notin N\} \\ H_0 &= \bigcup_{n \geq 0} G_n \\ H_{n+1} &= \{\Pi(v : B)x | B \in F \wedge x \in H_n \wedge t \notin B\} \\ \mathbb{P} &= \bigcup_{n \geq 0} H_n. \end{aligned}$$

◇ Forms of Constructor of  $t$  is generated by

$$\begin{aligned} G_0 &= \{t\} \\ G_{n+1} &= \{(xN) | x \in G_n \wedge N \in F \wedge t \notin N\} \\ H_0 &= \bigcup_{n \geq 0} G_n \\ H_{n+1} &= \{\Pi(v : B)x | B \in F \wedge x \in H_n \wedge [(t \notin B) \vee (t \in B \wedge v \notin x)]\} \\ \mathbb{C} &= \bigcup_{n \geq 0} H_n. \end{aligned}$$

**Definition 1.5.2** Given  $t, Q : \Pi(a : \mathbf{A})\mathcal{U}$ , which is an arity, we can define a function  $F_{t|Q}^n : \mathbb{C}(t) \rightarrow F$  by induction:

$$\begin{aligned} F_{t|Q}^n(t) &= Q \\ \text{if } F_{t|Q}^n(x) &\text{ is defined} \\ F_{t|Q}^n(xN) &= F_{t|Q}^n(x)N \\ \text{if } B \in F &\text{ and } t \notin B, \\ F_{t|Q}^n(\Pi(v : B)x) &= \Pi(v : B)F_{t|Q}^n(x) \\ \text{if } B \in \mathbb{P}(t) &\text{ and } t \in B, \\ F_{t|Q}^n(\Pi(v : B)x) &= B \rightarrow B \xrightarrow{Q}_t \rightarrow F_{t|Q}^n(x). \end{aligned}$$

**Definition 1.5.3** Let  $t : \Pi(\mathbf{a} : \mathbf{A})\mathcal{U}$ ,  $Q : \Pi(\mathbf{a} : \mathbf{A})(ta) \rightarrow \mathcal{U}$ , and  $c \in \mathbb{C}(t)$ , the function  $F_{t|Q}^d : \mathbb{F} \times \mathbb{C} \rightarrow \mathbb{F}$  is given by:

$$\begin{aligned}
& F_{t|Q}^d(r, t\mathbf{M}) = (Q\mathbf{M})r \\
& \text{if } F_{t|Q}^d(\_, x) \text{ is defined, for all } r \in \mathbb{F} \\
& \text{if } B \in \mathbb{F} \text{ and } t \notin B, \\
& \quad F_{t|Q}^d(r, \Pi(v : B)x) = \Pi(v : B)F_{t|Q}^d((rv), x) \\
& \text{if } \mathbb{P}(t) \ni B = \Pi(\mathbf{b} : \mathbf{B})(t\mathbf{N}) \text{ and } t \in B, \\
& \quad F_{t|Q}^d(r, \Pi(v : B)x) = \Pi(v : B)[\Pi(\mathbf{b} : \mathbf{B})((Q\mathbf{N})(v\mathbf{b})) \rightarrow F_{t|Q}^d((rv), x)].
\end{aligned}$$

**Definition 1.5.4** Let  $t : \Pi(\mathbf{a} : \mathbf{A})\mathcal{U}$ ,  $f \in \mathbb{F}$ , the function  $F_{t|f}^e : \mathbb{F} \times \mathbb{C}(t) \rightarrow \mathbb{F}$  is given by

$$\begin{aligned}
& F_{t|f}^e(r, t) = r \\
& \text{if } F_{t|f}^e(\_, x) \text{ is defined, for all } r \in \mathbb{F} \\
& \quad F_{t|f}^e(r, xN) = F_{t|f}^e(r, x) \\
& \text{if } B \in \mathbb{F} \text{ and } t \notin B, \\
& \quad F_{t|f}^e(r, \Pi(v : B)x) = \lambda(v : B)F_{t|f}^e((rv), x) \\
& \text{if } \mathbb{P}(t) \ni B = \Pi(\mathbf{b} : \mathbf{B})(t\mathbf{N}) \text{ and } t \in B, \\
& \quad F_{t|f}^e(r, \Pi(v : B)x) = \lambda(v : B)F_{t|f}^e(rv\lambda(\mathbf{b} : \mathbf{B})(f\mathbf{B}(v\mathbf{b})), x).
\end{aligned}$$

### CIC: Form

Without losing generality, suppose that  $X \in V$ ,  $C_i \in \mathbb{C}(X)$  and  $\Pi(\mathbf{a} : \mathbf{A})\text{Type}_0 \in \mathbb{A}$ , then

$$\begin{aligned}
& D \quad \Gamma \quad X : \Pi(\mathbf{a} : \mathbf{A})\text{Type}_0 \quad C_0 : \text{Type}_0 \\
& \quad \vdots \\
& D \quad \Gamma \quad X : \Pi(\mathbf{a} : \mathbf{A})\text{Type}_0 \quad C_m : \text{Type}_0 \\
& D \quad \Gamma \quad \mathbf{X} : \Pi(\mathbf{a} : \mathbf{A})\text{Type}_0
\end{aligned}$$

### CIC: Constr

$$\begin{aligned}
& D \quad \Gamma \quad \mathbf{X} : \Pi(\mathbf{a} : \mathbf{A})\text{Type}_0 \\
& D \quad \Gamma \quad (\text{Constr}\mathbf{X}) : C_i \frac{\mathbf{X}}{\mathbf{X}}
\end{aligned}$$

### CIC: Elim-1

$$\begin{aligned}
& D \quad \Gamma \quad \mu : (\mathbf{X}\mathbf{W}) \\
& D \quad \Gamma \quad Q : \Pi(\mathbf{a} : \mathbf{A})\text{Type}_0 \\
& D \quad \Gamma \quad f_i : [F_{X|Q}^n(C_i)] \frac{\mathbf{X}}{\mathbf{X}} \\
& D \quad \Gamma \quad (\text{Ind}\mathbf{X})(Qf_0 \cdots f_m\mu) : (Q\mathbf{W})
\end{aligned}$$

### CIC: Elim-2

$$\begin{aligned}
& D \quad \Gamma \quad r_i : C_i \\
& D \quad \Gamma \quad \mu : (\mathbf{X}\mathbf{W}) \\
& D \quad \Gamma \quad Q : \Pi(\mathbf{a} : \mathbf{A})(X\mathbf{a}) \rightarrow \text{Type}_0 \\
& D \quad \Gamma \quad f_i : [F_{X|Q}^d(r_i, C_i)] \frac{\mathbf{X} \text{ Constr}\mathbf{X}}{\mathbf{X} \quad r_i} \\
& D \quad \Gamma \quad (\text{Ind}\mathbf{X})(Qf_0 \cdots f_m\mu) : (Q\mathbf{W})\mu
\end{aligned}$$

**CIC: Compu**

For any  $Q, f_i, M_i, A_i \in \mathbf{F}$ ,  $\mathbf{X}, \mathbf{i} \in C$  and  $a_i, \mu \in V$ ,

$$(\mathbf{IndX})(Qf_0 \cdots f_m(\mathbf{ConstrXiM})) \rightarrow_{\mathbf{X}} F_{\mathbf{X}|\lambda(\mathbf{a}:A)\lambda(\mu:(\mathbf{Xa}))}^e \mathbf{IndX}Qf_0 \cdots f_m\mu(f_i, M)$$

and Compatible  $\rightarrow_{\mathbf{X}}$ .